A Additional Dataset and Experimental Setup Details

Long-Tail Datasets Balanced datasets can be artificially transformed into long-tail (LT) datasets by assigning class sample counts according to a geometric progression governed by an imbalance ratio ρ. In this formulation, the head class has the maximum number of samples, N, while the tail class has approximately ρN samples. The number of samples in class i is given by:

$$n_i = \left\lfloor N \rho^{\frac{i}{C-1}} \right\rfloor \tag{8}$$

where N is the number of samples in the head class, ρ is the imbalance ratio ($0 < \rho < 1$), $i \in \{0, 1, \ldots, C-1\}$ is the class index, and C is the total number of classes.

CIFAR10-LT and CIFAR100-LT The original CIFAR10 and CIFAR100 datasets each consist of a training set with 50k images uniformly distributed across 10 or 100 classes, respectively. Their long-tailed variants, CIFAR10-LT and CIFAR100-LT [27], introduce an exponential decay in class frequency from class 0 to the final class. Common long-tail imbalance ratios include $\rho = 0.01$ and $\rho = 0.001$. Specifically for $\rho = 0.01$, CIFAR10-LT contains 12,406 images, with the first head class comprising 5,000 samples and the last tail class only 50. CIFAR100-LT has 10,847 images, with the head class containing 500 samples and the tail just 5.

Experiments on CIFAR10-LT and CIFAR100-LT were conducted using NVIDIA A100 80GB SXM GPUs. Training took approximately 7 hours, and sampling required 8 hours. For DDPM, CBDM, and CORAL, the hyperparameters used were: a learning rate of 2×10^{-4}, batch size of 128, Adam optimizer with default momentum parameters, dropout rate of 0.1, 150k training steps, and $T = 1000$ diffusion steps. For T2H, all settings remained the same except for the number of training steps, which was increased to 200k.

CelebA-5 CelebA-5 [16] is a five-class subset of the CelebA dataset, composed of samples labeled with exactly one of the following hair colors: black, brown, blonde, gray, or bald. Samples with multiple or missing labels are excluded. The dataset is naturally imbalanced, with black- and brown-haired individuals significantly outnumbering those with gray hair or baldness.

Experiments on CelebA-5 were run on NVIDIA H100 GPUs. Training took approximately 18 hours, and sampling required 22 hours. All models were trained with a learning rate of 3×10^{-4} and a batch size of 128, with all remaining hyperparameters kept consistent with the CIFAR experiments.

Hyperparameters We summarize the regularization hyperparameters and sampling guidance scale ω used for each method and dataset. The sub-tables correspond to DDPM (top left), CBDM (top right), T2H (bottom left), and CORAL (bottom right), respectively. For CORAL, the base contrastive weight, w, in (7) was set to 0.01.

DDPM [2]

Dataset	ω
CIFAR10-LT ($\rho = 0.01$)	0.8
CIFAR10-LT ($\rho = 0.001$)	1.0
CIFAR100-LT ($\rho = 0.01$)	0.8
CelebA-5	0.6

CBDM [5]

Dataset	ω	τ_{cb}
CIFAR10-LT ($\rho = 0.01$)	1.0	1.0
CIFAR10-LT ($\rho = 0.001$)	1.8	1.0
CIFAR100-LT ($\rho = 0.01$)	1.6	1.0
CelebA-5	1.0	50.0

T2H [6]

Dataset	ω
CIFAR10-LT ($\rho = 0.01$)	1.0
CIFAR10-LT ($\rho = 0.001$)	1.7
CIFAR100-LT ($\rho = 0.01$)	1.5
CelebA-5	1.0

CORAL (ours)

Dataset	ω	τ_{SC}	τ_r
CIFAR10-LT ($\rho = 0.01$)	0.6	0.12	0.8
CIFAR10-LT ($\rho = 0.001$)	1.0	0.10	1.0
CIFAR100-LT ($\rho = 0.01$)	0.8	0.09	1.0
CelebA-5	0.7	0.12	0.8

Table 3: Hyperparameter settings for each method.

We follow the code implementation for CBDM [5] with the regularization weight $\tau_{\mathrm{cb}} = \tau/T$, where τ is the original weight defined in [5], and T is the total number of diffusion timesteps. For T2H [6], we use the code implementation with direct distance-based weighting that does not require a regularization parameter.

B Additional Results

Generated Images We generate images for CIFAR10-LT, CIFAR100-LT, and CelebA-5 to evaluate the effectiveness of CORAL in addressing the challenges of diffusion models trained on long-tailed datasets. Randomly selected examples are shown in Figures 5 to 7, illustrating how our contrastive latent alignment framework improves both the quality and diversity of generated samples, particularly for tail classes.

Figures 5 and 6 show generated samples for CIFAR10-LT and CIFAR100-LT, respectively, with $\rho = 0.01$. CORAL successfully disentangles latent representations to generate high-quality, diverse samples for the underrepresented classes. CORAL preserves the distinctive characteristics of each class, effectively mitigating feature borrowing from head to tail classes. The visual quality of these results highlights the effectiveness of CORALs latent space regularization in promoting class separation and maintaining clean, well-structured features in the generated outputs.

Figure 7 displays generated samples for CelebA-5, demonstrating CORALs ability to handle naturally imbalanced data. The dataset exhibits pronounced class imbalance across five hair color categories (black, brown, blond, gray, and bald) with the head class containing nearly 15 times more samples than the tail class. In such imbalanced settings, latent representations for tail classes often become entangled with those of head classes. CORAL effectively preserves class-specific features, producing diverse and realistic images in all categories.

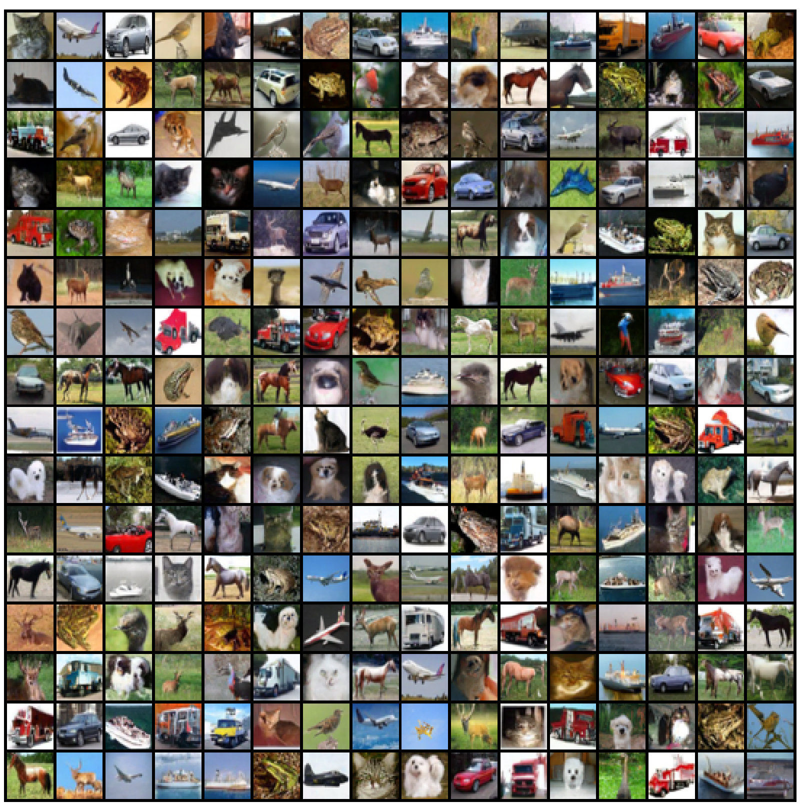

Figure 5: **Generated samples** produced by CORAL on the CIFAR10-LT dataset with $\rho = 0.01$.

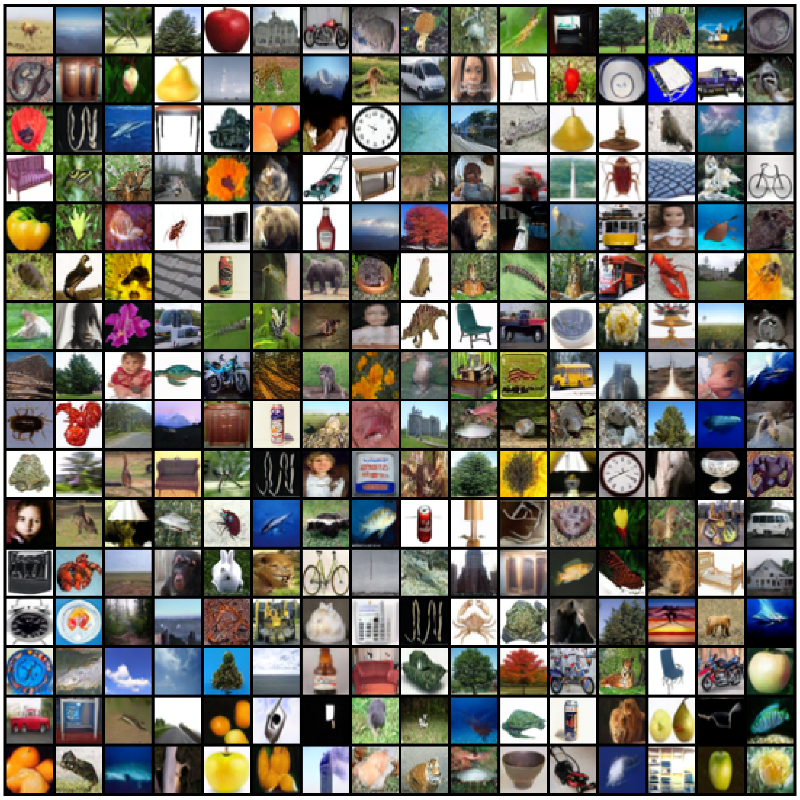

Figure 6: **Generated samples** produced by CORAL on the CIFAR100-LT dataset with $\rho = 0.01$.

Figure 7: **Generated samples** produced by CORAL on the CelebA-5 dataset.

C Ablation Studies

Effects of Hyperparameters Figure 8 illustrates the effect of three key hyperparameters on CORALs performance for CIFAR10-LT with imbalance ratio $\rho = 0.01$, measured by FID. The supervised contrastive temperature, τ_{SC}, in (5) achieves optimal performance at $\tau_{\text{SC}} = 0.12$, beyond which FID increases sharply.

For the decay rate temperature, τ_r, in (7), the best performance is observed at $\tau_r = 0.8$. FID remains relatively stable for $0.7 \leq \tau_r \leq 0.9$, with a steep increase outside of this range. These findings support our hypothesis that contrastive regularization is most effective when applied toward the end of the denoising process (i.e., when $t \sim 0$).

Finally, for the CFG scale, ω, in (4), FID traces a convex curve with optimal performance at $\omega = 0.6$. This indicates a trade-off between leveraging class-conditional information ($\omega > 0$) and avoiding over-conditioning that could limit sample diversity ($\omega \gg 0.6$). These results underscore the importance of careful hyperparameter tuning in CORAL to achieve an optimal balance between sample fidelity and diversity, particularly for long-tailed datasets.

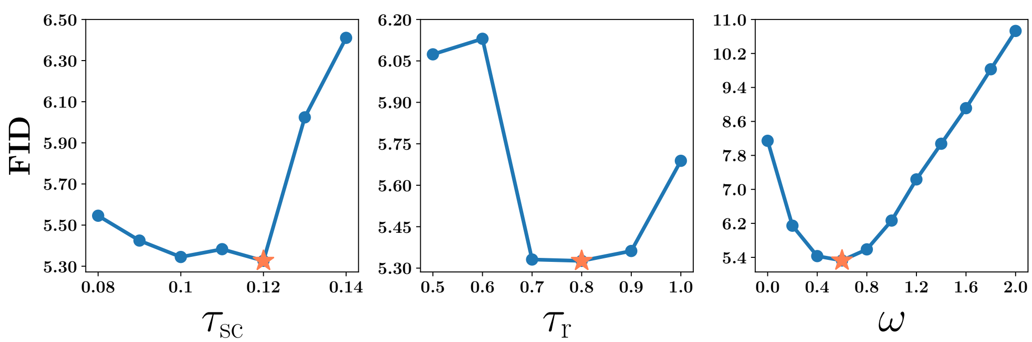

Figure 8: **Effect of regularization hyperparameters and guidance scales on FID.** From left to right: FID vs τ_{SC}, FID vs τ_r, and FID vs ω for the CIFAR10-LT dataset. Coral stars mark the lowest FID achieved for each hyperparameter.

CORAL on Balanced Datasets We also evaluate CORAL on two balanced datasets, CIFAR10 and CIFAR100, using FID as the performance metric. Even in the absence of class imbalance, CORAL outperforms DDPM and CBDM in terms of FID. This improvement stems from CORALs contrastive loss, which promotes class-wise separation in the latent space even in balanced settings, as illustrated in Figure 9. Importantly, this separation is achieved without compromising fidelity or diversity, as reflected in the consistently strong FID scores.

Dataset	Method	FID (↓)
CIFAR10	DDPM [2]	3.84
	CBDM [5]	3.61
	CORAL (ours)	**3.30**
CIFAR100	DDPM [2]	3.91
	CBDM [5]	3.37
	CORAL (ours)	**2.86**

Table 4: Comparison of methods on CIFAR10 and CIFAR100 image generation.

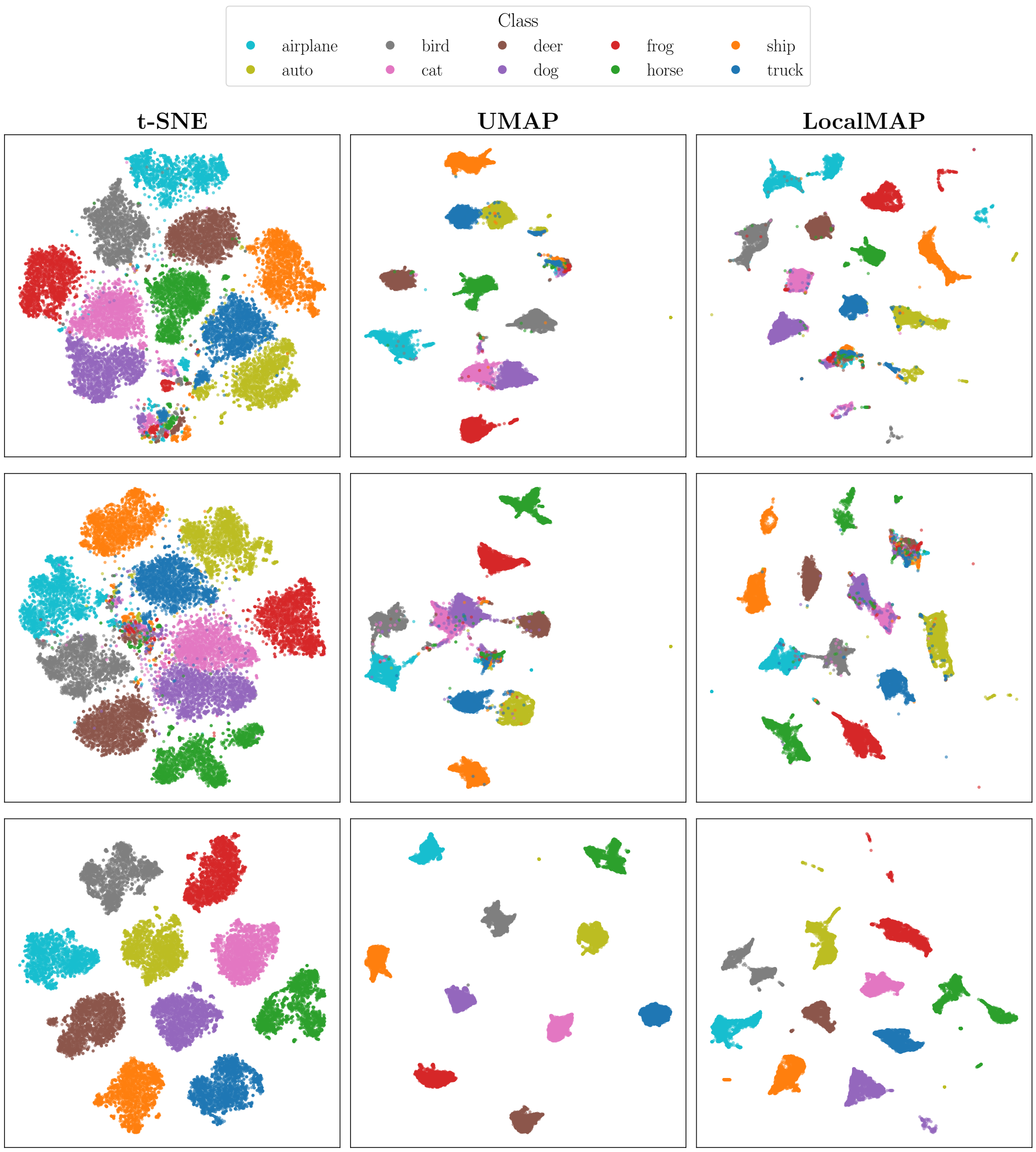

Figure 9: **Visualizations of U-Net bottleneck features** using t-SNE, UMAP, and LocalMAP on CIFAR-10 (balanced). Each row shows a different method trained on CIFAR-10, listed **from top to bottom**: DDPM [18], CBDM [5], and CORAL (ours).

D Comparison with Other Methods

Latent Space Visualization Figure 10 visualizes the latent representations from the U-Net bottleneck layer using t-SNE [11] (left), UMAP [34] (middle), and LocalMAP [35] (right) for DDPM [18], CBDM [5], T2H [6], and CORAL trained on CIFAR10-LT with an imbalance ratio of $\rho = 0.01$. CORAL exhibits markedly improved class-wise separation in latent space, mitigating the representational entanglement that typically causes feature mixing between head and tail classes. Figure 11 presents analogous visualizations for CelebA-5 across all methods except T2H, for which no implementation is available on this dataset.

Comparison with Baseline Methods We compare CORAL with baseline methods and highlight its strengths to address long-tailed generation in diffusion models.

- **CBDM:** Introduces a distribution adjustment regularizer during training that encourages similarity between generated images across different classes, transferring knowledge from head classes to tail classes. CBDM [5] suffers from mode collapse because its regularization loss encourages the model to produce similar outputs across different class conditions.
- **T2H:** Employs weighted denoising score matching to transfer knowledge from head classes to tail classes by using head samples as denoising targets for noisy tail samples. Its performance depends on both label distribution and sample similarity. T2H's [6] score substitution mechanism could potentially lead to mode collapse when noisy tail samples are consistently mapped to the same limited set of head references due to similarity-based selection.
- **CORAL:** As demonstrated in our experimental results, CORAL consistently outperforms both CBDM and T2H across a range of evaluation metrics, with particularly strong gains in tail classes. Our experimental results across multiple datasets highlight the strengths of CORAL:
  1. **Mode Stability**: CORAL prevents mode collapse, and generates class-consistent and visually diverse samples. As can be seen in the generated samples. This is in contrast to CBDM, which often fails to preserve class identity, *e.g.,* by generating class-conditioned samples displaying attributes of other classes, as shown in Figure 4.

     CORAL effectively reduces undesirable interclass feature borrowing in the class labeled datasets we consider. At the same time, CORAL allows the transfer of non-discriminative features that facilitate generalization, as shown in Figure 4.
  2. **Adaptive Regularization**: CORAL incorporates time-dependent regularization into the contrastive loss. This adaptive weighting enhances separation during the later stages of denoising, when outputs are less noisy and more semantically meaningful. Figure 8 shows that contrastive regularization is most effective when applied toward the end of the denoising process.
  3. **Latent Disentanglement**: CORALs strength lies in leveraging the lower-dimensional latent space of the denoising U-Net, which has been shown to capture semantically meaningful structure [10]. CORAL achieves effective inter-class disentanglement in the latent space by employing a linear projection head (see Figure 2), resulting in high-fidelity and class-aligned generated samples. These effects are illustrated in Figure 10 for CIFAR10-LT with $\rho = 0.01$ and in Figure 11 for CelebA-5, using latent space visualizations from t-SNE, UMAP, and LocalMAP.

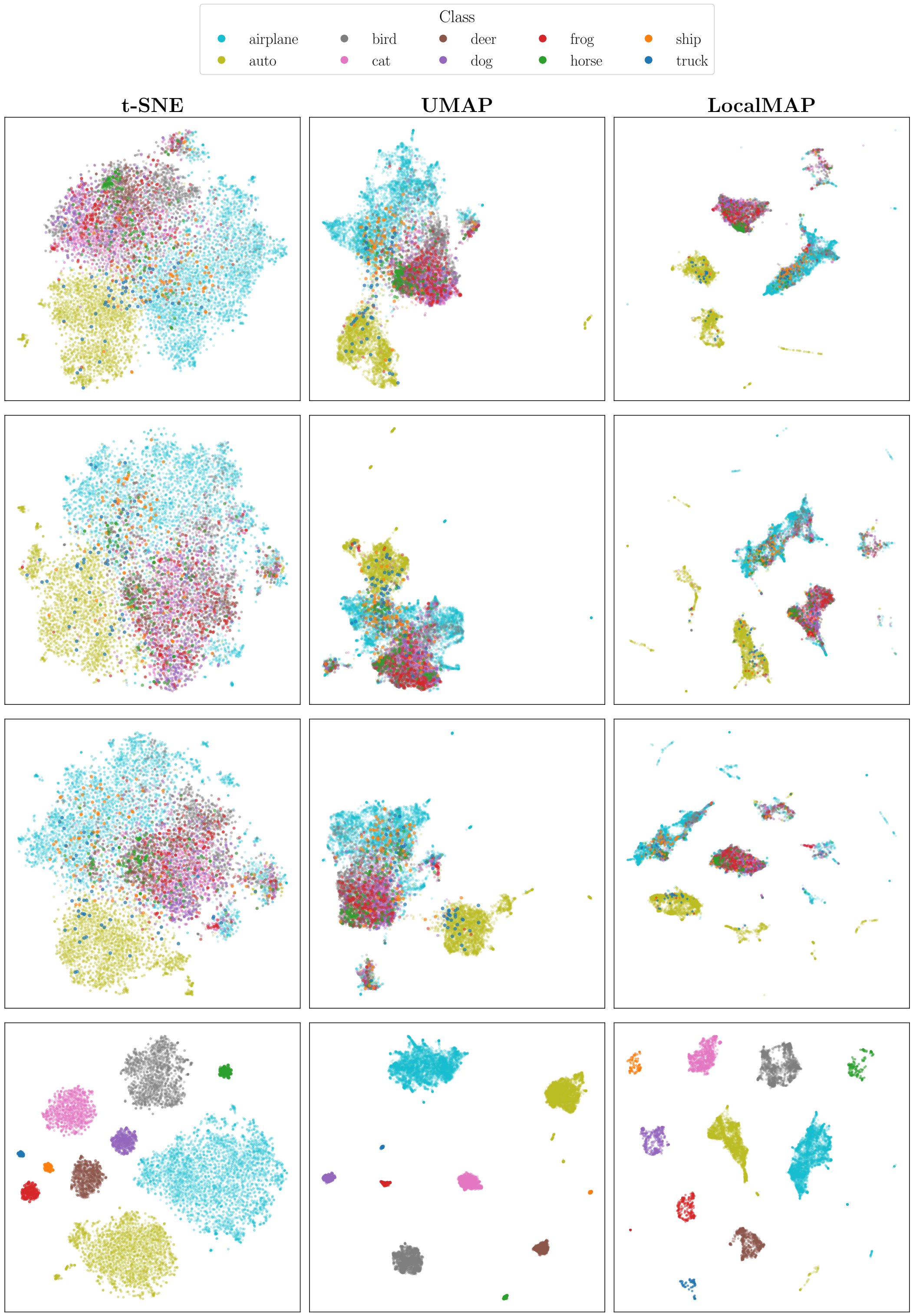

Figure 10: **Visualizations of U-Net bottleneck features** using t-SNE, UMAP, and LocalMAP on CIFAR10-LT with an imbalance ratio of $\rho = 0.01$. Each row shows a different method trained on CIFAR10-LT, listed **from top to bottom**: DDPM [18], CBDM [5], T2H [6], and CORAL.

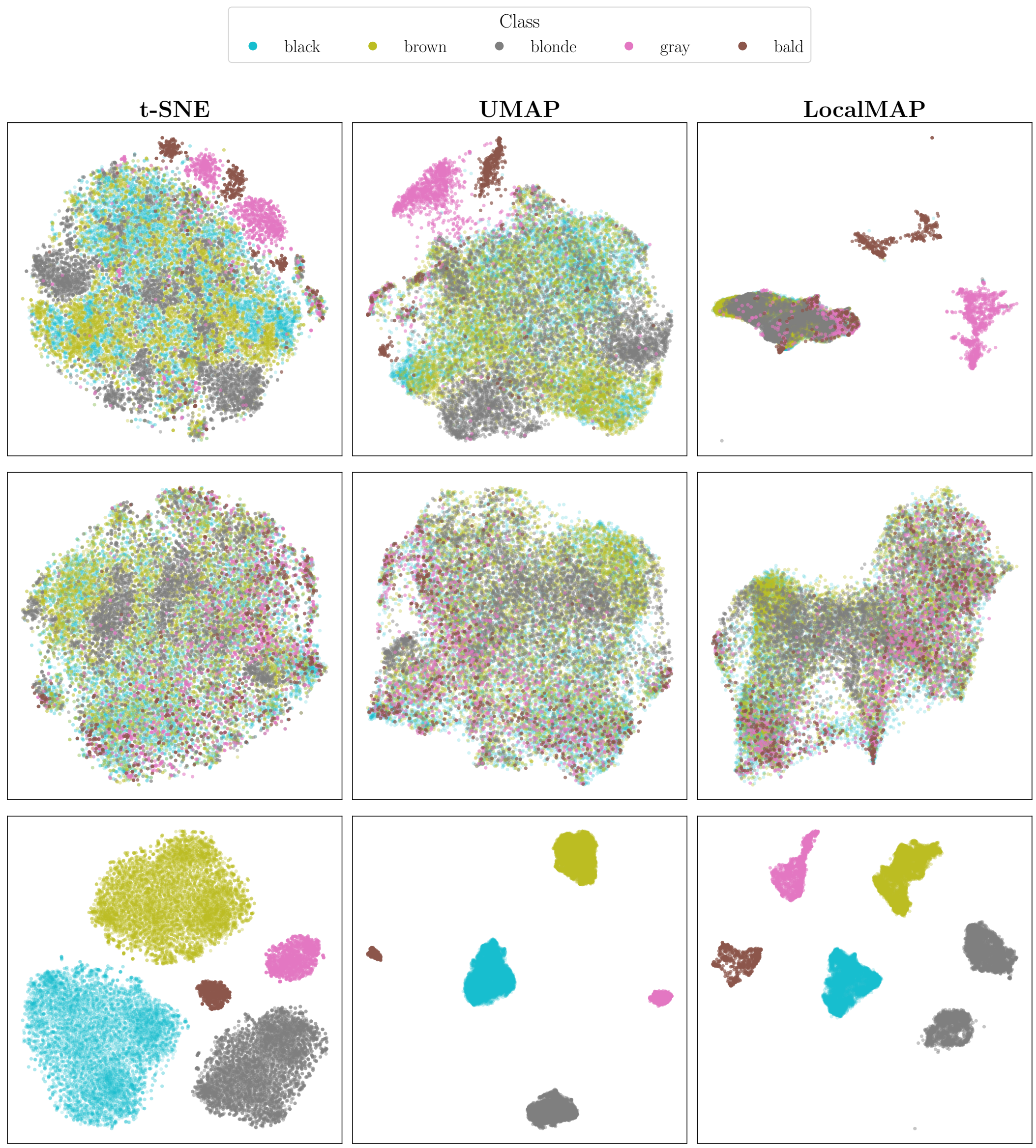

Figure 11: **Visualizations of U-Net bottleneck features** using t-SNE, UMAP, and LocalMAP on CelebA-5. Each row shows a different method trained on CelebA-5, listed **from top to bottom**: DDPM [18], CBDM [5], and CORAL.